Retest · Algebra I Module 2 Topic 1

Answer Key · Forms A, B, C (teacher copy)

---

1 point each. Item 1 matches the original retest item 1 on every form, item 2 matches item 2, and so on.

Item	Question	Answer	Work
A1	What is the point of the line given by the point-slope equation $y - 5 = \frac{2}{3}(x + 1)$? Enter your answer as an ordered pair.	**(−1, 5)**	$y - y_1 = m(x - x_1)$: $x_1 = -1$, $y_1 = 5$.
A2	Rewrite the equation in slope-intercept form. Simplify your answer. **$y - 2 = 3(x - 4)$**	**y = 3x − 10**	y − 2 = 3x − 12 → y = 3x − 10
A3	What is the equation in slope-intercept form of a line with a slope of $\frac{1}{3}$ that passes through the point (−3, 2)?	**y = (1/3)x + 3**	2 = (1/3)(−3) + b → 2 = −1 + b → b = 3
A4	What is the slope of the line given by the equation $2x + 5y = 10$?	**−2/5**	5y = −2x + 10 → y = (−2/5)x + 2
B1	What is the point of the line given by the point-slope equation $y + 2 = \frac{5}{4}(x - 8)$? Enter your answer as an ordered pair.	**(8, −2)**	y + 2 = y − (−2): $x_1 = 8$, $y_1 = -2$.
B2	Rewrite the equation in slope-intercept form. Simplify your answer. **$y + 5 = -2(x - 1)$**	**y = −2x − 3**	y + 5 = −2x + 2 → y = −2x − 3
B3	What is the equation in slope-intercept form of a line with a slope of $\frac{3}{4}$ that passes through the point (4, −1)?	**y = (3/4)x − 4**	−1 = (3/4)(4) + b → −1 = 3 + b → b = −4
B4	What is the slope of the line given by the equation $4x - 2y = 8$?	**2**	−2y = −4x + 8 → y = 2x − 4

C1	What is the point of the line given by the point-slope equation $y - 7 = -\frac{1}{2}(x + 3)$? Enter your answer as an ordered pair.	**(–3, 7)**	$x + 3 = x - (-3)$: $x_1 = -3$, $y_1 = 7$.
C2	Rewrite the equation in slope-intercept form. Simplify your answer. **$y - 6 = 5(x + 2)$**	**y = 5x + 16**	$y - 6 = 5x + 10 \rightarrow y = 5x + 16$
C3	What is the equation in slope-intercept form of a line with a slope of $-\frac{2}{5}$ that passes through the point (5, 1)?	**y = –(2/5)x + 3**	$1 = (-2/5)(5) + b \rightarrow 1 = -2 + b \rightarrow b = 3$
C4	What is the slope of the line given by the equation $6x + 3y = 9$?	**–2**	$3y = -6x + 9 \rightarrow y = -2x + 3$